Linear Algebra

Matrix

Oct.20th - Oct.22nd, 2026 Lecture

simultaneous equations

Simultaneous equations are a finite set of equations for which common solutions are sought. **system of linear equations**, one of those is a collection of two or more linear equations involving the same variables. For instance,

$$\begin{cases} 3x + 5y - z = 1 \\ -2x + 3y + 4z = 8 \\ 37x - 2y + z = -3 \end{cases}$$

* Please do not try to solve this It's mathematical gibberish

Systems of linear equations are generally solved by using the substitution method or the elimination method to remove specific variables —effectively canceling out matching terms— thereby simplifying the equations to find the solution.

Example 1:

Solve systems of linear equations below:

$$\begin{cases} 2x - y = 1 \\ 3x + y = 9 \end{cases}$$

Sol)

$$3x + y = 9 \rightarrow y = 9 - 3x \cdots \text{a}$$

$$Substitute\ \text{a}\ to\ \ 2x - y = 1. Then,$$

$$2x - 9 + 3x = 5x - 9 = 1, \qquad \therefore x = 2$$

$$2 - y = 1, \qquad \therefore y = 1$$

$$\therefore \boldsymbol{x = 2, y = 1}$$

Invention of Matrix

Matrices were invented to efficiently solve systems of linear equations and to systematically express geometric transformations (like rotation and scaling)

Assume that there's nth degrees simultaneous equation likes below:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = s_1 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = s_n \end{cases}$$

By stripping away the variables and the equal signs, we can rewrite this system as a single matrix equation:

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} s_1 \\ \vdots \\ s_n \end{bmatrix}, \qquad AX = b$$

It can also be represented by the equation shown form, typically expressed using coefficient matrix A, vector X, and constant matrix b.

Or simply,

$$\left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & s_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & s_n \end{array}\right]$$

The matrix of this form is called an "augmented matrix" and is primarily used in the context of Gaussian elimination. The line separating the coefficients from the solutions represents the equal sign.

Exercise 1:

Express given simultaneous equations to the matrix:

$$\begin{cases} 2x - y = 1 \\ 3x + y = 9 \end{cases}$$

Introduction to Matrix

Matrixes are usually described with their size. The size of a matrix is defined by its **dimensions**, written as $m \times n$, read as "m by n"

Structure of Matrix

- **Rows (n)**: The **horizontal** lines of numbers.
- **Columns (m)**: The **vertical** lines of numbers.

$$\begin{matrix} \uparrow \\ m \\ \downarrow \end{matrix} \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad \begin{matrix} \\ \\ \leftarrow n \rightarrow \end{matrix}$$

In addition, **Square Matrix** is a matrix where the number of rows equals the number of columns (i.e. $m = n$)

Regardless of its dimensions, a matrix in which all elements are zero is called a zero matrix and is denoted by the symbol O.

$$O = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}$$

Example:

Determine dimension of matrix below

$$\begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 0 \end{bmatrix}$$

Sol)

$$m = 2, n = 3$$

$\therefore \mathbf{2 \times 3}$

Matrix addition is a **component-wise** (or elementwise) operation. It is only defined when two matrices have the exact same dimensions.

Techniques – Addition of Matrixes

Addition of matrix only defined under condition of matrixes have same dimensions

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} + \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & \cdots & a_{nn} + b_{nn} \end{bmatrix}$$

Or simply

$$A + B = C$$

Example 1:

Determine whether two pairs of matrixes given are addible or not.
Add both matrixes if pair is addible, if not, describe why.

A) $\begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 6 & 2 \\ 43 & 3 & 7 \end{bmatrix}$

B) $\begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 5 \\ 45 & 35 \\ 5 & 65 \end{bmatrix}$

Sol)

$$A)\ sum\ of\ 2 \times 2\ matrixes. Thus, both\ are\ addible.$$

$$\therefore \begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 6 & 2 \\ 43 & 3 & 7 \end{bmatrix} = \begin{bmatrix} 7 & 10 & 3 \\ 45 & 8 & 7 \end{bmatrix}$$

$$B)\ sum\ of\ 2 \times 2\ and\ 3 \times 2\ matrixes. Addition\ can\ not\ defined.$$

Matrix multiplication (AB) is **not** elementwise. It is defined based on the inner product (dot product) of rows and columns.

Techniques – Multiplication of Matrixes

If number of columns in the first matrix equals the number of rows in the second matrix, then both matrixes are multipliable to each.

To find the entry in the i-th row and j-th column of the product, you take the dot product of the i-th row of First one and the j-th column of Second one

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \times \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} (a_{11} \times b_{11}) + \cdots + (a_{1n} \times b_{n1}) & \cdots & (a_{11} \times b_{1n}) + \cdots + (a_{1n} \times b_{nn}) \\ \vdots & \ddots & \vdots \\ (a_{n1} \times b_{11}) + \cdots + (a_{nn} \times b_{n1}) & \cdots & (a_{n1} \times b_{1n}) + \cdots + (a_{nn} \times b_{nn}) \end{bmatrix}$$

Or simply

$$A \times B = C$$

The product of an $m \times n$ matrix and an $n \times l$ matrix results in an $m \times l$ matrix.
Square Matrixes are always multipliable. For instance,

$$\begin{bmatrix} 2 & 3 \\ 6 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} (2 \times 1) + (3 \times 4) & (2 \times 3) + (3 \times 2) \\ (6 \times 1) + (4 \times 4) & (6 \times 3) + (4 \times 2) \end{bmatrix} = \begin{bmatrix} 14 & 12 \\ 22 & 16 \end{bmatrix}$$

If number of columns in the first matrix equals the number of rows in the second matrix, then both matrixes are multipliable to each.

For example, **(2×2) and (2×3)**, **(4×3) and (3×8)** are multipliable.
But **(2×5) and (2×3)**, **(5×6) and (8×6)** are not.

Matrix multiplication is non-commutative, even when you multiply multiple matrices.
For three matrix A, B and C, generally ABC, multifaction of three matrixes is equal to A(BC) and (AB)C, but not equal to BCA or CBA.

$$ABC = A(BC) = (AB)C$$

$$ABC \neq BCA \neq CAB$$

I.e. matrix multiplication is associative but not non-commutative.
However, repetitive multiplication of same matrixes is associative and commutative. It described as **"exponentiation of matrix"**

$$A^n = A^{n-k}A^k = A^kA^{n-k}$$

Another exception where the distributive law applies—or rather, a specific case involving it—is the product of a matrix and an **identity matrix**. An identity matrix is a matrix that, when multiplied by any other matrix from either side, results in the original matrix; it is defined as follows:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+0 & b+0 \\ 0+c & 0+d \end{bmatrix} = A$$

Multiplication of Matrixes – Commutativity

Matrix multiplication is directional, and the commutative law does not hold.
Thus,

$$ABC \neq BCA \neq CAB$$

As an exception, powers of the same matrix and multifaction with Identity matrix are commuted with each other.

$$A^n = A^{n-k}A^k = A^kA^{n-k}$$
$$IA = AI = A$$

Since the distributive law holds in some cases but not in others, caution is required when performing the following operations.

$$A(B+C) = AB + AC$$

$$(B+C)A = BA + CA$$

In addition, while operation of matrixes differs much than calculation of simple numbers, there exists a quality that should not be overlooked.

Multiplication of Matrixes – Zero divisor

A zero divisor is a non-zero matrix that, when multiplied by another non-zero matrix, yields a zero matrix. i.e.

$$AB = O$$
$$both\ A, B\ \neq O$$

Suppose matrix A and matrix B are both non-zero matrices. If we compute the product $\boldsymbol{AB}$, every entry becomes zero, resulting in the zero-matrix $\boldsymbol{O}$. For instance,

$$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}\begin{bmatrix}0 & 0\\2 & 3\end{bmatrix} = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix} = O$$

This also means that the cancellation law does not hold, a point that should be kept in mind when performing calculations such as the following.

$$AB - AC = A(B-C) = O$$

$$Thus, AB = AC\ not\ implies\ B = C$$

Determinants

In general, solutions for the system of linear equation below:

$$\begin{cases} ax + by = s_1 \\ cx + dy = s_2 \end{cases}$$

Described as below:

$$x = \frac{s_1 d - b s_2}{ad - bc}$$

$$y = \frac{a s_2 - s_1 c}{ad - bc}$$

* Those are simply derived by making one of two equations to equation for single variable x or y...

The solutions above all have $ad - bc$ in the denominator, and the existence of a solution depends on the value of $ad - bc$. Naturally, if this expression equals zero, no solution exists. We call this the **determinant**.

This also applies to matrixes, where it is used to represent their state, characteristics, and the like. It is expressed as shown below.

Determinants of 2×2 matrixes

For 2×2 matrix **A** below:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Its determinant is described as below:

$$\det A = \begin{vmatrix} d & b \\ c & d \end{vmatrix} = ad - bc$$

The **determinant** is undoubtedly one of the most fundamental and vital concepts in linear algebra. It acts as a single-number summary that reveals the core properties of a matrix.

Determinants of n×n matrix

In the lecture, we cover determinants for matrices up to size 3x3. The approach for square matrices of higher dimensions follows the same logic: the determinant is calculated by reducing the dimension down to 2x2.

When the dimension of a determinant is reduced to two, the even-indexed determinants within the overall sequence of determinants must be negative.

e.g. Determinants of 3×3 matrixes

For 3×3 matrix **A** below:

$$A = \begin{bmatrix} i & j & k \\ a & b & c \\ d & e & f \end{bmatrix}$$

Its determinant is described as below:

$$\det A = i \begin{vmatrix} b & c \\ e & f \end{vmatrix} - j \begin{vmatrix} a & c \\ d & f \end{vmatrix} + k \begin{vmatrix} a & b \\ d & e \end{vmatrix}$$

To solving simultaneous equations via matrix – Cramer's rule

To find solutions of system of linear equation below:

$$x = \frac{s_1 d - b s_2}{ad - bc}, \qquad y = \frac{a s_2 - s_1 c}{ad - bc}$$

Cramer's rules are often used.

For the simultaneous equation below:

$$\begin{cases} ax + by = s_1 \\ cx + dy = s_2 \end{cases}$$

And it's matrix form

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} \rightarrow Ax = b$$

And if the column vector $\begin{bmatrix} x \\ y \end{bmatrix}$ is the solution of Ax = b, then

$$x_i \det A = \det A_i$$

Holds true. If determinant of A is not 0, we can use Cramer's rule defined as below.

Cramer's rule

For $\det A \neq 0$

$$x_i \det A = \det A_i$$

$$\therefore \; x_i = \frac{\det A_i}{\det A}$$

Here, A_i is the matrix obtained by replacing the ith column of A with s.

To form A_i

For the matrix equation

$$Ax = b \rightarrow \begin{bmatrix} a_{11} & \cdots & a_{1i} \\ \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ii} \end{bmatrix} \begin{bmatrix} x \\ \vdots \\ x_i \end{bmatrix} = \begin{bmatrix} s_1 \\ \vdots \\ s_i \end{bmatrix}$$

$$A_1 = \begin{bmatrix} s_1 & \cdots & a_{1i} \\ \vdots & \ddots & \vdots \\ s_i & \cdots & a_{ii} \end{bmatrix}$$

$$A_2 = \begin{bmatrix} a_{11} & s_1 & \cdots \\ \vdots & \vdots & \vdots \\ a_{i1} & s_i & \cdots \end{bmatrix}$$

$$\vdots$$

$$A_i = \begin{bmatrix} a_{11} & \cdots & s_1 \\ \vdots & \ddots & \vdots \\ a_{11} & \cdots & s_i \end{bmatrix}$$

To change ith column to column vector b